

1. Administrative & Study Identification

Full Study Title: A Multicentre, Randomised, Open-Label, Parallel-Group, Phase IIb Study to Assess the Potential for Tezepelumab-treated Patients With Severe Asthma to Reduce Background Therapy While Sustaining Asthma Control and Clinical Remission **Short Title/Acronym:** ARRIVAL Study **Protocol Number:** D5180C00047 **NCT Number:** NCT06473779 **Sponsor Name:** AstraZeneca **Investigational Product:** Tezepelumab (TEZSPIRE®) 210 mg **Phase of Development:** Phase IIb **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the potential for tezepelumab-treated patients (subcutaneous administration) to reduce maintenance background therapy without the loss of asthma control at Week 56, among those who demonstrated asthma control or low biomarkers at Week 24. **Secondary Objectives:** To evaluate the proportion of patients who achieve asthma control and have characteristics of clinical and biological remission at Week 24, as well as to monitor changes in T2 inflammatory biomarkers (FeNO and blood eosinophils) and lung function.

2.2 Background & Rationale To address the gap in long-term severe asthma management regarding the necessity of lifelong background inhaled corticosteroids (ICS) and other maintenance therapies. With the broad anti-T2 inflammatory effects of tezepelumab, this study evaluates whether patients can safely step down their background asthma medications while sustaining clinical remission, thereby reducing the cumulative side effects of chronic corticosteroid exposure.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Parallel-group (Active Standard of Care Continuation vs. Reduction) **Blinding:** Open-label **Randomization:** Randomised

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 326 evaluable patients **Number of Sites:** ~65 sites across 10 countries (including Europe, Canada, US, and RoW) **Estimated Study Start:** 30-Sep-2024 **Estimated Primary Completion:** 25-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients 12 to 80 years of age inclusive at the time of consent. 2. Documented physician-diagnosed severe asthma within 10 years prior to Visit 1. 3. Asthma Control Questionnaire (ACQ-5) score ≥ 1.5 and < 3 with a documented history of at least one asthma exacerbation requiring OCS bursts or hospitalization within 12 months prior to screening.

4.2 Exclusion Criteria 1. Any clinically important pulmonary disease other than asthma (e.g., COPD, cystic fibrosis, bronchiectasis). 2. Current smokers, or patients with a smoking history ≥ 10 pack-years, including current users of vaping products/e-cigarettes. 3. A helminth parasitic infection diagnosed within 6 months prior to Visit 1 or major surgery within 8 weeks prior to Visit 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tezepelumab 210 mg administered once every 4 weeks (Q4W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Up to 68 weeks of treatment (total study duration up to 72 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Additional background maintenance asthma controller medications (e.g., LTRAs, tiotropium, cromone, theophylline) prior to the reduction phase. **Prohibited:** Concurrent initiation of other monoclonal antibodies or biologic therapies targeting T2 inflammation during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening/Run-in	Week -4 to 0	Informed Consent, Medical History, Inhaler technique check, Baseline ACQ-5 & Biomarkers
Baseline	Week 0	Randomization, First Dose of Tezepelumab 210mg
Interim Assessment	Week 24	Efficacy Assessment (Asthma Control/Biomarkers), Decision to initiate background therapy reduction
Maintenance/Observation	Q4W	Subcutaneous dosing, Safety Labs, Symptom Questionnaires
Primary Endpoint Eval	Week 56	Assessment of sustained asthma control vs. loss of control post-reduction
End of Study	Week 68-72	Final dosing (Wk 68), Final Vital Signs, Exit Interview, Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of patients who successfully reduce background maintenance therapy without experiencing a loss of asthma control at Week 56 (assessed among those with controlled asthma at Week 24). **Measurement Method:** Clinical exacerbation monitoring and ACQ-5 threshold scoring.

7.2 Secondary & Exploratory Endpoints * Proportion of patients achieving clinical remission criteria at Week 24. * Changes in fractional exhaled nitric oxide (FeNO) and blood eosinophil counts (BEC). * Changes in pre-bronchodilator \$FEV_1\$ (lung function stability).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of adverse events at every Q4W clinical visit and via patient daily symptom diaries. **Serious Adverse Events (SAEs):** Reported to the sponsor/regulatory bodies within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External independent Data Safety Monitoring Board (DSMB) to assess ongoing risk-benefit profiles throughout the background therapy reduction phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE incidence. **Sample Size Justification:** \$N=326\$ sufficiently powered (\$\alpha = 0.05\$, \$Power = 80\%\$) to demonstrate the non-inferiority/superiority of maintaining clinical remission following background therapy withdrawal. **Handling of Missing Data:** Multiple Imputation for missing ACQ-5 scores; patients missing Week 56 data may be treated as "loss of control" or LOCF depending on the reason for dropout.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Q4W onsite visits combined with centralized risk-based remote monitoring. **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking, automated data clarification forms (DCFs), and investigator site file audits.

11. Study Identifiers & Governance

Primary ID: D5180C00047 **Registry Number:** NCT06473779 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARRIVAL **Official**

Regulatory Title: A Multicentre, Randomised, Open-Label, Parallel-Group, Phase IIIB Study to Assess the Potential for Tezepelumab-treated Patients With Severe Asthma to Reduce Background Therapy While Sustaining Asthma Control and Clinical Remission.

12. Clinical Framework (Severe Asthma Specific)

Primary Condition: Severe Uncontrolled Asthma **Diagnosis Threshold:** ACQ-5 ≥ 1.5 and < 3 and history of severe exacerbations **Medical Methodology:** Biologic therapy targeting TSLP (Tezepelumab) combined with protocolized withdrawal of background controller medication **Optimization Target:** Sustained Asthma Control and Clinical Remission (absence of exacerbations, stable lung function)

13. Study Procedures & Methodology

Management Pathway: Standardized step-down algorithm for inhaled corticosteroids and background controllers **Education Protocol (HCP):** Training on safe medication withdrawal thresholds and exacerbation action plans **Education Protocol (Patient):** Inhaler technique training (mandatory prior to baseline) and self-monitoring of worsening symptoms **Quality Audit Mechanism:** Real-time tracking of ACQ-5 scores, pulmonary function tests, and biomarker (FeNO/Eos) stabilization

14. Observation & Follow-Up Schedule

Total Duration: 72 Follow-up weeks **Onsite Visit Cadence:** Weeks 0, 24, 56, 68, 72 (with interim checks) **Remote Monitoring Frequency:** Standard Q4W touchpoints concurrent with SC injections **Checklist Review Interval:** Continuous daily diary assessment reviewed comprehensively every 4 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				